Huawei Access Node BAS-ACC-37 - Routing Protocol Flap Failure SOP (ALARM_2082)

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2082

Description

This Standard Operating Procedure (SOP) addresses a critical failure scenario on Huawei's Access Node BAS-ACC-37 at Basavanagudi POP, precipitated by a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding). The underlying protocol mechanics responsible for this failure domain involve the dynamic interaction between the Routing Information Bases (RIBs) of the BGP, OSPF, and ISIS routing protocols, as they strive to maintain a consistent and up-to-date view of the network topology. In the event of a failure, these protocols may engage in a rapid exchange of routing information (Advertising, withdrawing, or re-advertising routes), resulting in a 'flap' of the routing system. This can lead to a cascade of events, including network instability, packet loss, and in extreme cases, complete network partitioning. The Huawei Access Node, specifically, may become overwhelmed by the high rate of routing updates, leading to a degradation in its ability to process and forward traffic.

Pre-requisite Checks

- Verify that the Basavanagudi POP site is accessible and that all necessary personnel and equipment are present.

- Confirm that the redundant link status is normal and that the backup path is functioning correctly.

- Ensure that all necessary safety procedures are in place and that all personnel are aware of the potential risks involved in the troubleshooting process.

Potential Root Causes

Cause	Probability
Hardware failure in the BGP routing engine or a related component	High
Software bugs or misconfigurations in the BGP routing protocol	Medium
Inadequate network design or planning	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: BGP routing table information, including the number of routes, the number of updates, and the number of withdrawals, Description: This command provides an overview of the BGP routing table, helping to identify whether the flap is caused by a specific route or a general problem with the routing protocol.

Command: display ospf routing-table, Expected Output: OSPF routing table information, including the number of routes, the number of
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updates, and the number of withdrawals, Description: This command
provides an overview of the OSPF routing table, helping to identify
whether the flap is caused by a specific route or a general problem
with the routing protocol.

Command: display isis routing-table, Expected Output: ISIS routing
table information, including the number of routes, the number of
updates, and the number of withdrawals, Description: This command
provides an overview of the ISIS routing table, helping to identify
whether the flap is caused by a specific route or a general problem
with the routing protocol.
```

Resolution Steps

Step 1: Step: Clear BGP routing table, Command: reset bgp all, Description: This command clears the BGP routing table, forcing the node to relearn the routing information from its neighbors. This can help to resolve the flap caused by a corrupted or inconsistent routing table.

Step 2: Step: Recalculate OSPF routing table, Command: ospf recalculate, Description: This command recalculate the OSPF routing table, helping to identify and resolve any issues that may be causing the flap.

Step 3: Step: Check and repair ISIS routing table, Command: isis check and repair, Description: This command checks the ISIS routing table for any issues and repairs any damaged or corrupted data.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: BGP routing table
information, including the number of routes, the number of updates,
and the number of withdrawals, Description: This command verifies that
the BGP routing table is stable and that there are no more flapping
routes.

Command: display ospf routing-table, Expected Output: OSPF routing
table information, including the number of routes, the number of
updates, and the number of withdrawals, Description: This command
verifies that the OSPF routing table is stable and that there are no
more flapping routes.

Command: display isis routing-table, Expected Output: ISIS routing
table information, including the number of routes, the number of
updates, and the number of withdrawals, Description: This command
verifies that the ISIS routing table is stable and that there are no
more flapping routes.
```

Roll-back Procedures

- Step: Restore the original BGP configuration, Command: rollback bgp config, Description: This command restores the original BGP configuration, reversing any changes made during the troubleshooting process.

- Step: Restore the original OSPF configuration, Command: rollback ospf config, Description: This command restores the original OSPF configuration, reversing any changes made during the troubleshooting process.

- Step: Restore the original ISIS configuration, Command: rollback isis config, Description: This command restores the original ISIS configuration, reversing any changes made during the troubleshooting process.